Basic Level Questions

1. Create Student Table

Create a `students` table with:

- student_id
- name
- age
- course
- marks

Conditions:

- `student_id` should be PRIMARY KEY
- `name` should NOT NULL
- `marks` should have default value 0

2. Insert Data

Insert 5 student records into the `students` table.

3. Update Query

Update marks of the student whose name is "Rahul" to 85.

4. Delete Query

Delete students whose age is less than 18.

5. Select Query Practice

Write queries to:

- Show students with marks greater than 70
- Display students ordered by age in descending order
- Show distinct courses
- Display first 3 characters of student names

Data Types Practice

6. Employee Table

Create a table with proper data types:

Column	Data Type
emp_id	INT
emp_name	VARCHAR
salary	DECIMAL
joining_date	DATE
is_active	BOOLEAN

7. Product Table

Choose appropriate data types for:

- Product name
- Price
- Quantity
- Manufacturing date
- Description

Constraints Practice

8. Users Table

Create a `users` table with:

Conditions:

- user_id → PRIMARY KEY
- email → UNIQUE
- age → must be greater than 18
- city → default value 'Lucknow'

9. NOT NULL Practice

Create a table where:

- username
- password

cannot contain NULL values.

10. CHECK Constraint

Create a table where salary can never be negative.

Primary Key & Foreign Key Practice

11. Department and Employee Tables

Create two tables:

departments

- dept_id
- dept_name

employees

- emp_id
- emp_name
- dept_id

Condition:

- dept_id should be a FOREIGN KEY.

12. One-to-Many Relationship

One customer can have multiple orders.

Design:

- customers table
- orders table

Use proper primary and foreign keys.

Medium Level Questions

13. School Database Design

Create tables:

- students
- teachers
- subjects
- student_subjects

Define relationships properly.

14. Library Management System

Design tables for:

- books
- students
- issued_books

Add constraints also.

15. E-Commerce Database

Create tables:

- users
- products
- orders
- order_items

Define all foreign keys.

Query Practice (Medium)

16. JOIN Query

Display employee and department data together.

17. INNER JOIN

Show only employees who belong to a valid department.

18. Aggregate Functions

Find:

- Maximum salary
- Average salary
- Total number of employees

19. GROUP BY

Count employees in each department.

20. HAVING Clause

Show only departments where employee count is greater than 3.

Slightly Advanced Questions

21. ON DELETE CASCADE

Use `ON DELETE CASCADE` with foreign keys.

Test:

- When parent record is deleted, child records should also be deleted.

22. Composite Primary Key

Create a `student_courses` table where:

- student_id
- course_id

together form a composite primary key.

23. Self Join Concept

Store employees and managers in the same table.

Hint:

- manager_id should reference the same table as a foreign key.

24. Multiple Constraints Practice

Create a banking table where:

- account_number is UNIQUE
- balance cannot be negative
- account_type default value should be 'saving'

Mini Project Challenge

25. Online Course Platform Database

Design a complete database with tables:

- students
- instructors
- courses
- enrollments
- payments

Requirements:

- Proper primary keys
- Foreign keys
- Constraints
- Appropriate data types
- Sample INSERT queries
- 5 useful SELECT queries

JOIN Practice Questions

1. INNER JOIN

Create:

- students
- courses
- enrollments

Write a query to show:

- student name
- course name
- enrollment date

using INNER JOIN.

2. LEFT JOIN

Show all students including those who are not enrolled in any course.

3. RIGHT JOIN

Show all courses even if no student enrolled.

4. Multiple JOINs

Create:

- customers
- orders
- products
- order_items

Display:

- customer name
- product name
- quantity
- order date

5. SELF JOIN

Create an employee table with:

- emp_id
- emp_name
- manager_id

Show employee name and manager name together.

Subquery Practice

6. Highest Salary

Find employees who have salary greater than average salary.

7. Second Highest Salary

Find the second highest salary from employees table.

8. Students Above Average Marks

Show students whose marks are greater than average marks.

9. EXISTS Subquery

Show customers who have placed at least one order.

10. IN Clause with Subquery

Show employees working in departments located in Delhi.

GROUP BY & HAVING

11. Department Wise Salary

Show:

- department name

- total salary
- average salary

using GROUP BY.

12. HAVING Clause

Show departments having more than 5 employees.

13. Product Sales

Find total quantity sold for each product.

14. Top Selling Products

Show products whose total sales are greater than 100.

LIMIT & OFFSET Practice

15. Pagination Query

Display:

- first 5 students
- next 5 students

using LIMIT and OFFSET.

16. Top 3 Highest Salaries

Show top 3 highest paid employees.

17. Lowest 2 Product Prices

Display 2 cheapest products.

Transactions Practice

18. Simple Transaction

Perform:

- money deduction from one account
- money addition to another account

using:

- START TRANSACTION
- COMMIT

19. ROLLBACK Practice

Insert records into a table and rollback the transaction.

Check whether data is saved or not.

20. Savepoint Practice

Use:

- SAVEPOINT
- ROLLBACK TO SAVEPOINT

inside a transaction.

Real-World Query Challenges

21. E-Commerce Query

Find customers who ordered more than 3 products.

22. Most Expensive Product

Find product(s) having maximum price.

23. Duplicate Records

Find duplicate emails from users table.

24. Monthly Sales Report

Show:

- month
- total orders
- total revenue

grouped month-wise.

25. Customers Without Orders

Find customers who never placed any order.

Advanced JOIN + Subquery Combined

26. Highest Paid Employee Per Department

Show highest salary employee from each department.

27. Latest Order of Each Customer

Display latest order details for every customer.

28. Employees Earning More Than Their Manager

Using SELF JOIN, find employees earning more than managers.

Constraints & Database Design

29. Banking Database

Design:

- accounts
- transactions
- customers

with:

- foreign keys
- unique constraints
- check constraints

30. Hospital Management System

Design tables:

- doctors
- patients
- appointments
- medicines

Add relationships and constraints.

Practice on Clauses

31. BETWEEN Clause

Show employees whose salary is between 30,000 and 60,000.

32. LIKE Clause

Find students whose names start with 'A'.

33. ORDER BY Multiple Columns

Sort employees:

- by department ascending
- then salary descending

34. DISTINCT Clause

Show distinct city names from customers table.

35. CASE Statement

Create a query:

- marks >= 90 → Excellent

- marks >= 70 → Good
- else → Average

Mini Project Challenges

36. Online Food Delivery Database

Design complete database for:

- restaurants
- users
- orders
- delivery partners
- payments

Write:

- joins
- subqueries
- reports

37. WhatsApp Chat Database

Design:

- users
- chats
- messages
- groups

Write queries for:

- latest messages
- unread messages
- group chats